

PIT WALL: Technical Verification of Granite TimeSeries TTM on Motorsport Telemetry

TL;DR

- **Verdict: novel.** No published paper, IBM cookbook recipe, Hugging Face Space, Kaggle notebook, watsonx tutorial, or hackathon project applies IBM Granite TimeSeries TTM (`ibm-granite/granite-timeseries-ttm-r2`) — or any other named time-series foundation model (Chronos, Chronos-2, TimesFM, MOIRAI, Moment, Lag-Llama) — to motorsport telemetry as of May 2026. The "Best Use of Technology" novelty claim is defensible.
 - **Strongest defensible task for a 12-day hackathon build:** few-shot fine-tuned multivariate forecasting of **per-corner lap-trace pace deltas and tyre/brake load proxies** for the *next* lap, conditioned on driver-control channels as **future-known exogenous** inputs. Demo at 1-Hz aggregation on a `512-96-r2` model — *not* raw 50-100 Hz signals, which TTM is not pretrained for.
 - **Recommendation:** Ship PIT WALL as a hackathon win and a NeurIPS Workshop on Time Series ("ML4Sports" or similar) short paper. Treat the spinout as plausible but second-order: TTM-on-adaptive-motorsport is a defensible thin-wedge product, but Track Titan (\$5M seed, Dec 2025, Partech + Game Changers Ventures co-leads, with 200,000+ users improving their fastest lap by more than half a second after their first session) already owns the able-bodied AI-coach niche. The differentiator is the FIA Certificate of Adaptations + Granite Guardian safety-envelope loop, not TTM itself. (`Tech.eu + 3`)
-

Key Findings

1. Novelty verdict: confirmed negative across all searched venues

A targeted multi-pass search (web_search + targeted subagent verification) returned **zero** prior art on Granite TimeSeries TTM applied to motorsport telemetry:

- **arXiv / Google Scholar / Semantic Scholar:** No paper combining "TinyTimeMixer" or "Granite TimeSeries" with motorsport, F1, racing, lap time, telemetry, FastF1, MoTeC, AiM, NASCAR, IndyCar, WEC, IMSA, GT3, rally, or karting. The only TTM paper that mentions any vehicular topic is the original Ekambaram et al. NeurIPS 2024 paper (arXiv 2401.03955), and its evaluation datasets are exclusively Monash + LOTSA (energy, weather, traffic, retail, M5) — no motorsport.
- **IBM Granite community repos:** `ibm-granite-community/granite-timeseries-cookbook` recipes cover energy demand (Spain), Kaggle bike-sharing, M5 retail sales, and TSPulse

classification on UEA BasicMotions. There is no racing/motorsport recipe. [\(ibm-granite-community/SageMaker/granite-timeseries-ttm-r2/sample_notebook.ipynb\)](#) uses generic numeric channels.

- **Hugging Face:** No motorsport-tuned TTM checkpoint or Space exists under [\(ibm-granite\)](#), [\(ibm-research\)](#), or community uploads as of May 2026.
- **Devpost / Kaggle / MLH archives:** No prior hackathon project pairs TTM with race telemetry.

Adjacent published work in motorsport ML (does not invalidate novelty — these use bespoke models, not TSFMs):

- *Explainable Time Series Prediction of Tyre Energy in Formula One Race Strategy* (arXiv 2501.04067) — uses three RNN variants + a Transformer encoder + linear/XGBoost baselines on FIA F1 telemetry, with TIME + CausalImpact for XAI. Forecasts tyre energies one step ahead from throttle/brake/steering covariates. **Does not use any TSFM.** [\(arXiv\)](#)
- *Data-driven pit stop decision support for Formula 1 using deep learning models* (Frontiers in AI, 2025; Bi-LSTM, TCN-GRU, GRU, InceptionTime, CNN-BiLSTM on FastF1 data). **Does not use any TSFM.** [\(Frontiers\)](#)
- *Formula RL: Deep RL for Autonomous Racing using Telemetry Data* (arXiv 2104.11106) — DDPG variants, not forecasting. [\(arXiv\)](#)
- Deakin University blog (2017) — generic survey of time-series modelling for motorsport; pre-foundation-model era.

Broader pattern (also negative): No published application of Chronos, Chronos-2, TimesFM, MOIRAI, Moment, Lag-Llama, Toto, or TabPFN-TS to motorsport telemetry was found.

Therefore the claim "**first application of a pretrained time-series foundation model to motorsport telemetry**" is currently defensible, not just "first TTM application." That is a stronger framing for the pitch deck.

2. What TTM r2 / r2.1 actually does — and does not — handle

From the official model card ([\(ibm-granite/granite-timeseries-ttm-r2\)](#)), READMEs at commits [\(b2a0242\)](#) and [\(d6a7957\)](#):

- **Sizes:** 1-5M parameters. Pretrained on ~700M (r2) and ~1B (r2.1) samples from Monash + LOTSA. [\(Hugging Face + 3\)](#)
- **Context lengths:** 512, 1024, **1536** (r2 only; r1 caps at 1024). [\(Medium\)](#) [\(Hugging Face\)](#)
- **Prediction lengths:** up to **720** (r2); 96 (r1). [\(Medium\)](#)
- **Resolutions supported:** minutely → hourly (10 min, 15 min, 1 h) for r2; r2.1 adds **daily and weekly**. There is **no native support for sub-second / 10-100 Hz sample rates**. This is the

single biggest constraint for telemetry. (Hugging Face) (Hugging Face)

- **Multivariate:** channel-independent by default; **decoder channel-mixing** (`decoder_mode="mix_channel"`) can be enabled during fine-tuning. (IBM)
- **Exogenous + future-known channels:** supported via `exogenous_channel_indices` and `prediction_channel_indices` in `TinyTimeMixerForPrediction.from_pretrained(...)`. A "control column" in IBM docs is an exogenous channel whose value is known into the future. (IBM)
- **Categorical infusion:** supported (useful for circuit ID, compound, weather state).
- **r2.1 "ft" variants:** *frequency prefix tuning* — an embedding vector indicating data frequency is prepended to the input; improves performance for short contexts (per IBM Granite 3.2 announcement).
- **Hard requirement:** users must externally standard-scale each channel before inference. `TimeSeriesPreprocessor` in `tsfm_public` handles this. (Hugging Face)
- **Documented limitations:** "Limited Resolution Support ... only supports minutely and hourly resolutions"; "TTM requires a minimum context length of 512 or 1024 ... Enabling any upsampling or prepending zeros to virtually increase the context length for shorter-length datasets is not recommended and will impact the model performance" (HF model card). (Hugging Face)
- **Speed:** TTMs are "lightweight and can be executed even on CPU-only machines." Parseable's 2025 observability benchmark logged the 805K-parameter r2 variant as competitive on commodity hardware for 1-minute granularity. (Hugging Face)

Implication for telemetry at 50 Hz: 6,000-row lap traces sit entirely *below* TTM's pretraining distribution. You must **downsample** / **aggregate** the raw 50 Hz log to TTM-friendly resolution (e.g. 1-second windows of mean/max/RMS per channel) *or* re-bin to per-mini-sector summary statistics. **Do not feed raw 50 Hz signals.**

3. Strongest defensible TTM-on-telemetry task for the demo

After mapping TTM's strengths against typical race-weekend data:

Target task: Forecast the **next-lap per-mini-sector (e.g. 50 m or 1-second-bin) summary tensor** — `[sector_mean_speed, sector_max_brake_pressure, sector_throttle_integral, sector_steering_RMS, sector_peak_lat_G]` — for **N future laps** given context of the last K laps.

Channel mapping (concrete recommendation):

Channel	Role	Notes
<code>time_in_session</code> (one-hot tyre compound, fuel-load decile, ambient temp bin)	future-known exogenous (<code>exogenous_channel_indices</code>)	Known a priori for any stint plan
<code>lap_number</code> , <code>sector_idx</code> , <code>circuit_id</code>	categorical infusion	Stable across forecast horizon
<code>mean_speed</code> , <code>peak_lat_G</code> , <code>peak_long_G</code>	target (<code>prediction_channel_indices</code>)	Pace and grip proxies
<code>throttle_integral</code> , <code>brake_integral</code> , <code>steering_RMS</code> , <code>gear_changes</code>	target (driver-effort proxies)	These are the levers PIT WALL coaches against
Hand-control lever channels (when present)	target or exogenous	For adaptive drivers — these are the "differentiator" channels

Use **context length 512** with **prediction length 96** (`(512-96-r2)` variant, branch `main` of `ibm-granite/granite-timeseries-ttm-r2`). At 1-Hz aggregation, this is ~8.5 minutes of history → ~1.6 minutes of forecast, i.e. roughly **5-7 laps of context predicting the next 1-2 laps**, which is the sweet spot for "what would the next lap look like" framing.

For "what if the driver changed adaptation X by Y%": This is **not a TTM use case in the strict forecasting sense**. TTM does not do counterfactual interventional inference. Two defensible workarounds:

- Sweep the exogenous channel:** Re-run TTM inference with the adaptation-state channel manually shifted (e.g., new hand-brake gain). TTM will produce a corresponding forecast under the assumption that the conditioning distribution still applies — adequate for "directional" coaching narrative, not for causal claims.
- Frame as a *contrast*, not a counterfactual:** Compare PIT WALL's TTM forecast against a Granite-4.1-8B-generated reference trace from a similar non-adapted lap. The debrief LLM articulates the gap. This is honest and demo-defensible.

The "next 30 laps" framing in the original task brief is **not advisable** with TTM r2 at 1-Hz: 30 laps × ~90 s = 2,700 timepoints, well past the 720-prediction-length cap. **Reframe to next 1-2 laps**, and ladder predictions iteratively if more horizon is needed.

4. Latency budget for the 60-second demo on commodity hardware

Step	Realistic latency	Notes
CSV ingest + <code>TimeSeriesPreprocessor</code> standard-scale	0.5-2 s	Pandas-bound; trivial for 6,000-row file
TTM r2 zero-shot inference (<code>512-96-r2</code>), ~1M params, batch size 1)	< 1 s	HF card: "inference times taking only a few seconds"; <code>Dataloop</code> Parseable observability benchmark logged the 805K-param TTM as competitive at 1-minute granularity on CPU
Granite-Docling 258M COA PDF parse	2-4 s	258M params, easily fits on CPU
Granite Vision 4.1 4B timing- sheet parse	3-6 s on RTX 4060; 10- 15 s on M2 CPU	Use the 4B Vision model in its tables/charts sweet spot
Granite 4.1 8B Instruct debrief (300-500 output tokens)	3-5 s on RTX 4060 at q4_K_M — SpecPicks (specpicks.com, Apr 2026): "8B is the sweet spot: 5GB VRAM at q4, ~75 tok/s on an RTX 4060"; <code>SpecPicks</code> 15- 30 s on M2 CPU at q4	Artificial Analysis (artificialanalysis.ai/models/granite-4-1- 8b) reports a provider P50 of 117.3 tokens per second <code>Artificial Analysis</code> — but that is hosted GPU infra, not laptop CPU
Granite Guardian 4.1 8B safety check	1-2 s on GPU; 5-10 s on CPU	Same 8B class. Run in parallel with debrief generation, not sequentially.
Langflow orchestration overhead	1-3 s	Visual graph, negligible if pre-warmed

Total demo budget at 60 s: Achievable on **RTX 4060 laptop only**. On M2 CPU, you will blow the budget at the debrief step. **Mitigation: pre-warm models, cache the COA parse, run Granite Guardian in parallel, and consider Granite 4.1 3B Instruct as a graceful CPU fallback** — though SpecPicks does not publish a per-card tok/s figure for the 3B variant, so test it empirically on Day 4.

Hard recommendation for the live demo: record a 60-second "hero" run on the RTX laptop, and have a *deterministic replay* mode (cached TTM forecast tensor + cached debrief) for the live

pitch. Live inference is for the Q&A.

5. Documented TTM r2.1 failure modes and how to engineer around them

Failure mode	Source	Mitigation
No support for sub-minute resolution	HF model card	Downsample 50 Hz telemetry to 1-Hz mini-sector aggregates
Minimum context length 512	HF model card	Use 5-7 laps $\times$ 90 s $\times$ 1 Hz to satisfy; never pad with zeros
External standard-scaling required per channel	HF model card; <code>tsfm_public.TimeSeriesPreprocessor</code>	Always run preprocessor; serialize the scaler with the demo artifact
Distribution mismatch with pretraining (energy/weather/retail, not telemetry)	TTM paper; IBM positions r1 vs r2 as "depends on target data distribution" <code>Hugging Face</code>	Few-shot fine-tune on 5-10 laps of real telemetry per circuit; IBM card claim: "easily be fine-tuned for multi-variate forecasts with just 5% of the training data" <code>Hugging Face +2</code>
Iterative multi-step rollouts compound error	General TSFM property	Stick to ≤ 1 forecast horizon (96 timepoints); do not rollout 30 laps
Channel-mixing requires fine-tuning, not zero-shot	HF card: "Decoder Channel-Mixing can be enabled during fine-tuning" <code>Hugging Face</code>	Plan ≥ 1 fine-tune pass; budget 10-20 min on a laptop
No native probabilistic outputs	TTM is point-forecast only	If demo needs uncertainty bands, ensemble across context lengths or use MC-dropout
No native categorical/text input	TTM card	Encode tyre compound, weather, circuit via the

Failure mode	Source	Mitigation
		exogenous numeric API or via the categorical infusion utility in tsfm_public
Pretraining excludes 10-100 Hz signals entirely	LOTS + Monash data composition	Communicate this clearly in the pitch: aggregation is a feature, not a bug — TTM forecasts <i>racecraft</i> , not control signals

6. Publication / IP / spinout assessment

Workshop paper: Yes — viable for NeurIPS Workshop on Time Series, ML4Sports (NeurIPS / ICML), or AAAI Workshop on Sports & AI. Framing: *"Zero-shot and few-shot foundation-model forecasting for adaptive motorsport: a multivariate TTM pipeline with safety-envelope governance."* Cite the gap explicitly — the F1 tyre-energy paper (arXiv 2501.04067) and the pit-stop Bi-LSTM paper (Frontiers AI 2025) both train bespoke models; PIT WALL is the first to bring a pretrained TSFM to the domain. The novelty bar at a workshop is low and your contribution clears it.

Patent: Plausible but weak as pure method. A method claim like "a multivariate time-series foundation model fine-tuned on adapted-vehicle control channels, gated by an FIA Certificate-of-Adaptations safety envelope, producing per-stint debrief narrative" *might* clear novelty (no clear prior art combines TSFM + adaptation-domain constraints + LLM debrief). However, the components are individually disclosed and the combination is straightforward; expect long prosecution. Better path: keep adaptation-domain priors and safe-envelope rules as **trade secret** + Apache-2.0 open-source TTM checkpoint as community asset.

Spinout / TAM: Trophi.ai retails at **\$12.99/month** Premium (sim-racing), with a Professional tier at **\$69.99/mo** that bundles a Driver61 1-on-1 session and on-track group classes. Track Titan raised **\$5M seed in Dec 2025** — co-led by Partech and Game Changers Ventures (Alpine F1 co-owner Roger Ehrenberg), with participation from Colton Parayko (Stanley Cup-winning NHL star), Trevoh Chalobah (Chelsea/England), Sequel (digital family office for athletes), On CEO Martin Hoffmann, Third Bridge founder Emmanuel Tahar, and existing investor APX (Axel Springer SE + Porsche AG). Per Tech.eu's coverage of the round (Dec 4, 2025): *"The platform has grown to more than 200,000 users and has recorded a tenfold increase in ARR over the past two years"*, with users "improving their fastest lap by more than half a second after their first session." MOZA + Fanatec hardware integrations are in production, and the explicit pitch is

"Strava for motorsport." The mainstream AI-coach market is *not* venture-greenfield anymore.

(SIMRACINGCOCKPIT + 12)

The defensible spinout wedge is adaptive motorsport specifically:

- The FIA's Disability and Accessibility Commission operates a Certificate of Adaptations process under Appendix L, Article 18.3 of the International Sporting Code. Disabled drivers must have COAs evaluated by the FIA Adaptations Working Group; non-ambulant drivers must affix a marshal-alert sticker. (FIA) (FIA)
- Existing organizations: Resilience Racing Foundation (US veterans), Team BRIT (UK), Spinal Track (UK), High Fives Foundation × Return to Dirt (motocross), MME Motorsports (Slovenia), SimAbility. (Resilienceracingfoundation)
- Population: per FIA Disability + Accessibility, "over 1 billion people in the world live with some form of disability"; the addressable adaptive-driver subset is small (low thousands of licensed competitors worldwide), but the *aspirational* funnel of disabled veterans + hobbyists is meaningfully larger and explicitly served by motorsport-therapy programs. (FIA)
- **Realistic TAM:** sub-\$50M ARR ceiling for AI-coach-tuned-for-adaptive-controls as a standalone SKU. Not venture-scale on its own. Venture-scale only if you treat adaptive motorsport as the *empathetic wedge* into a broader Trophi/Track Titan-style product — and at that point you're competing head-on with funded incumbents.

Verdict: Workshop paper > hackathon-win narrative > standalone spinout. The spinout angle is real but should be positioned as a non-profit / foundation partnership (with FIA Disability and Accessibility Commission, Resilience Racing Foundation, Team BRIT) rather than a venture pitch. Lead with the Best Use of Technology and Most Innovative claims; talk to FIA-DAA after the hackathon.

7. Adjacent IBM Granite tools the team may have missed

- **Granite 4.1 family (released April 29, 2026 per Artificial Analysis):** 3B, 8B, 30B dense models, plus Granite 4.1 Vision 4B, Granite Speech 4.1 2B NAR. The team is already using 8B Instruct + 4B Vision. **Consider Granite 4.1 3B Instruct as a CPU fallback** for the live demo — empirically benchmark on Day 4 because no public per-card tok/s figure is published for the 3B variant. (Artificial Analysis + 3)
- **Granite Embedding (incl. Granite-Embedding-30M-Sparse):** Sparse English embedding model released alongside Granite 3.2. Useful if PIT WALL wants RAG over COA PDFs or stewardly motorsport rulebooks (Appendix L, FIA International Sporting Code). (IBM)
- **Granite Code models:** Not directly relevant to PIT WALL's demo; do not add for the sake of adding.
- **watsonx.governance:** Overlaps with Granite Guardian for content-policy classification but is platform-grade. For a 12-day hackathon, **stay with Granite Guardian 4.1 8B**; mention

watsonx.governance as the production deployment story.

- **watsonx.ai Timeseries Forecasting API** (ibmdotcom-tutorials/generative-ai/time-series-forecasting-api.ipynb): Hosted endpoint for `ibm/granite-ttm-512-96-r2` with a simple `forecast()` call. Useful as a *fallback path* if local TTM inference flakes during the demo — same model, same outputs. (GitHub)
- **TSPulse (released alongside TTM r2.1)**: Time-series classification / anomaly detection / imputation companion model. **Strong candidate for an extension recipe**: classify "lap quality" (clean / dirty / off / spin) from telemetry, anomaly-detect mechanical failure precursors. Add as a stretch goal if the core demo is solid by day 8. (Hugging Face)

No motorsport-specific IBM partnership has been publicly announced in 2025–2026 (IBM's high-profile sports work centers on the US Open / tennis and the Masters / golf, not racing).

8. The June FIFA World Cup extension

Direct soccer translation of PIT WALL's TTM angle is feasible and equally novel.

- **No published TSFM (TTM, Chronos, Chronos-2, TimesFM, MOIRAI, Moment, Lag-Llama) application to soccer player tracking, GPS, biometric, or event data exists.** Adjacent prior work: a custom-trained soccer transformer (*A Foundation Model for Soccer*, arXiv 2407.14558) operates on discrete event-action sequences and is not a pretrained TSFM — verbatim: "*We propose a foundation model for soccer, which is able to predict subsequent actions in a soccer match from a given input sequence of actions... we train a transformer architecture on three seasons of data from a professional soccer league.*" The Royal Society Open Science 2025 paper on continuous player tracking from broadcast data (arXiv 2311.14642) frames position forecasting as a time-series problem but uses bespoke models. A 2025 bioRxiv paper (10.1101/2025.09.18.677041) on rugby explicitly motivates multivariate TSFMs: "*These advances motivate our use of multivariate time series models to capture the effect of collective player movement on performance prediction in Rugby Union tracking data.*" (arxiv) (biorxiv)
- **Natural June extension: TTM applied to GPS/biometric training-load + injury-risk forecasting** for World Cup squads. Data sources: Catapult, STATSports, Polar wearables (10–25 Hz). Multivariate channels: speed, acceleration, PlayerLoad, high-speed-running distance, HR, HRV. Target: forecast next-session load / injury risk per player. This re-uses PIT WALL's exact architecture pattern (multivariate TTM + Granite Guardian safe-envelope + Granite 4.1 debrief in the voice of a strength coach) and inherits the same novelty story.
- **Even better differentiator for June**: keep the *accessibility* thesis. There is a growing women's / parasports football audience, and the World Cup theme aligns naturally with adaptive football (CP football, blind football). The "Pit Wall for the Pitch" rebrand would be: an agentic post-match coach for grassroots / adaptive / women's football clubs, ingesting GPS + match event data, producing accessible debriefs.

This gives the team a coherent **two-month Grand Prize arc**: PIT WALL (May) → "TouchLine" (June, working name) — same Granite stack, same TTM-as-hero pattern, two different sports verticals, same accessibility-first thesis. Judges reward that kind of narrative consistency.

Details — Specific code-level recommendations

Install pin

```
bash

pip install "granite-tsfm[notebooks] @ git+https://github.com/ibm-granite/granite-tsfm"
```

(`v0.2.22`) is the version Alain Airom's IBM walkthrough verifies; latest minor may be newer — pin for reproducibility.) `Medium`

Model load — zero-shot first, fine-tune second

```
python

from tsfm_public.toolkit.get_model import get_model
from tsfm_public import TimeSeriesPreprocessor

TTM_MODEL_PATH = "ibm-granite/granite-timeseries-ttm-r2"
CONTEXT_LENGTH = 512          # 5-7 laps at 1 Hz aggregation
PREDICTION_LENGTH = 96       # ~1.5 laps forward

zeroshot_model = get_model(
    model_path=TTM_MODEL_PATH,
    context_length=CONTEXT_LENGTH,
    prediction_length=PREDICTION_LENGTH,
    freq_prefix_tuning=True,    # r2.1 frequency-tuned variants
    freq="s",                  # 1-second aggregation
)
```

Channel configuration

```
python
```

```

from tsfm_public.models.tinytimemixer import TinyTimeMixerForPrediction

target_columns = ["mean_speed", "throttle_int", "brake_int",
                  "steer_rms", "peak_lat_G", "peak_long_G"]
exogenous_columns = ["fuel_decile", "tyre_compound_id",
                    "ambient_temp_bin", "session_minute"]

tsp = TimeSeriesPreprocessor(
    timestamp_column="t",
    id_columns=["driver_id", "stint_id"],
    target_columns=target_columns,
    control_columns=exogenous_columns,    # known-future
    scaling=True,
    encode_categorical=True,
)

model = TinyTimeMixerForPrediction.from_pretrained(
    TTM_MODEL_PATH,
    num_input_channels=tsp.num_input_channels,
    prediction_channel_indices=tsp.prediction_channel_indices,
    exogenous_channel_indices=tsp.exogenous_channel_indices,
    decoder_mode="mix_channel",          # for the fine-tuned pass
)

```

Standard-scaling reminder

Per the model card: *"Users have to externally standard scale their data independently for every channel before feeding it to the model."* `TimeSeriesPreprocessor(scaling=True)` handles this; verify with `tsp.scaler_dict` before serializing for the demo. [Hugging Face](#) [Toolify](#)

Fine-tune protocol

```
python
```

```

for param in model.backbone.parameters():
    param.requires_grad = False # freeze backbone, train head
finetune_trainer = Trainer(
    model=model, args=finetune_args,
    train_dataset=dset_train, eval_dataset=dset_val,
    callbacks=[EarlyStoppingCallback(early_stopping_patience=3),
               TrackingCallback()],
    optimizers=(optimizer, scheduler),
)
finetune_trainer.train()

```

Budget: 10-20 minutes on a 4060-class GPU for 5-10 laps of fine-tune data; minutes on CPU.

Fallback / production path

If local inference flakes during the demo, swap to the **watsonx.ai Timeseries Forecasting API** (`ibm/granite-ttm-512-96-r2` model id, hosted) — same forecast values, network-bound latency ~2-4 s.

Recommendations (staged, with explicit triggers)

- By Day 4:** Validate end-to-end TTM inference on a real telemetry CSV (FastF1 export or a Resilience Racing / Team BRIT donated set). Threshold to ship: zero-shot MAE on next-lap `mean_speed` < classical seasonal-naive baseline. *If TTM does not beat seasonal-naive zero-shot, pivot immediately to fine-tune-first.* This is the single highest-risk gate. Also empirically benchmark Granite 4.1 3B vs 8B on the target laptop on the same day.
- By Day 6:** Lock channel mapping (Section 3 table). Freeze the demo's input schema; everything downstream depends on this.
- By Day 8:** Have the cached / replayable demo path working. Live inference only for Q&A.
- By Day 10:** Granite Guardian safe-envelope rules drafted from a real (or representative public) COA — e.g., max throttle gain, max steering deflection, hand-brake force ceiling. Guardian classifies the *forecast* against the envelope, not the historical lap.
- By Day 12:** Pitch deck — lead with *Best Use of Technology* (cite the negative-result novelty in this report) and *Most Innovative* (accessibility-first framing). Recommend not pitching the spinout aggressively; pitch the workshop-paper angle and the FIA Disability and Accessibility Commission collaboration.
- Post-hackathon (decision triggers):**
 - Workshop paper** if the demo works on real adaptive-driver data (even from a single driver). Submit by NeurIPS Time Series workshop deadline (typically Sept).

- **Spinout exploration** only if (a) FIA-DAA or Resilience Racing Foundation expresses formal interest, AND (b) at least 50 adaptive drivers commit to a paid pilot in the first 90 days. Otherwise, keep PIT WALL open-source.
 - **Pivot to June Grand Prize** with the GPS/biometric soccer reuse described in Section 8; lock that decision by June 3 to leave ≥ 3 weeks for build.
-

Caveats

- All performance benchmarks cited (Granite 4.1 8B at 117.3 tok/s provider P50 per Artificial Analysis; ~75 tok/s on an RTX 4060 per SpecPicks; ~11 tok/s on Raspberry Pi 5 + Hailo-8) are *vendor-reported* or *third-party benchmarks*. On-laptop latency for a *cold* model is dominated by load time, not steady-state generation. Pre-warm everything before the demo. (Artificial Analysis) (SpecPicks)
- The "novelty" verdict is based on exhaustive but not literally complete coverage. There is some residual probability that an unindexed internal IBM Research note, a paywalled industry journal (Racecar Engineering, RaceTech), or an undisclosed sim-racing-startup project uses TTM on telemetry. In the pitch, hedge with "to the best of our knowledge after a structured search of arXiv, Google Scholar, Semantic Scholar, Hugging Face, GitHub, Devpost, and Kaggle, no prior application of a time-series foundation model to motorsport telemetry has been published."
- TTM "supports daily and weekly" in r2.1 — your race-weekend cadence is *per-lap*, not per-day. Use r2 (512-96-r2) (minutely/hourly), not the r2.1 daily/weekly variants. Frequency-prefix-tuned r2.1 variants for hourly *may* outperform r2 with short context; A/B test by Day 4. (Hugging Face)
- The Track Titan figures (200,000+ users; 10 $\times$ ARR growth over two years; >0.5 s fastest-lap improvement after first session) are publicly reported via the company's Dec 4, 2025 funding announcement and Tech.eu's coverage; they are not independently audited. (Tech.eu)
- Trophi.ai's pricing has multiple SKUs (\$12.99 Premium / \$19.99 / \$69.99 Professional monthly tiers); the "\$7.50–\$12.99" figure in the original brief understates the higher tier. Use \$12.99 (Premium) as the relevant retail benchmark and \$69.99 (Professional) as the upper-bound human-coach-bundled SKU.
- For the *June* World Cup extension, GPS / wearable data access is non-trivial — Catapult, STATSports, etc. do not publish raw datasets at scale. Plan to use a public proxy (e.g., a parkrun / Strava-style GPS export, or the publicly available portion of StatsBomb 360 spatial data) for the demo.